

CHARLEVILLE-MEZIERES AP, France

WMO: 070750

Lat: 49.7828N

Lon: 4.6431E

Elev: 489

StdP: 14.44

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.2	21.4	11.6	10.1	22.1	15.8	12.4	23.7	21.5	46.1	18.7	44.7	1.7	300	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	85.5	68.9	81.9	67.4	78.7	65.8	70.8	82.5	68.8	79.4	67.0	76.4	6.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	99.5	76.4	64.8	93.6	73.7	63.3	88.7	71.6	34.8	82.6	33.2	79.5	31.7	76.8	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.9	14.8	13.2	DB	11.0	91.6	5.0	4.3	7.4	94.7	4.5	97.2	1.7	99.7	-2.0	102.8	(4)
			WB	10.6	74.8	4.9	2.3	7.1	76.4	4.2	77.7	1.5	79.0	-2.0	80.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	36.3	37.6	42.6	48.0	54.8	60.7	64.2	63.1	56.9	50.6	43.0	37.7	(6)
(7)		DBStd	11.63	7.64	6.80	5.98	6.13	5.70	5.79	5.62	5.17	5.06	6.40	6.71	7.38	(7)
(8)		HDD50	1822	425	348	236	110	18	1	0	0	5	71	225	383	(8)
(9)		HDD65	5717	889	768	693	511	318	154	86	99	246	447	661	845	(9)
(10)		CDD50	1710	1	0	7	49	168	322	439	405	213	89	14	3	(10)
(11)		CDD65	132	0	0	0	0	3	25	61	39	4	0	0	0	(11)
(12)		CDH74	1986	0	0	0	24	104	418	774	568	94	4	0	0	(12)
(13)		CDH80	532	0	0	0	1	13	105	231	170	12	0	0	0	(13)
(14)	Wind	WSAvg	5.6	6.5	6.4	6.4	5.7	5.4	5.2	5.0	4.4	4.5	5.2	5.8	6.3	(14)
(15)	Precipitation	PrecAvg	41.3	4.6	3.5	3.0	2.6	2.8	3.2	3.3	2.9	3.7	3.6	4.8	(15)	
(16)		PrecMax	50.4	11.1	8.2	7.1	7.2	5.9	5.4	7.3	6.8	6.1	8.4	6.4	11.7	(16)
(17)		PrecMin	30.7	0.7	0.7	0.2	0.4	0.4	1.0	1.6	1.0	0.9	1.6	0.7	1.1	(17)
(18)		PrecStd	5.5	2.2	1.9	1.7	1.4	1.5	1.5	1.3	1.9	1.4	1.5	1.6	2.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	57.1	67.2	77.4	81.7	88.2	91.4	91.2	81.6	73.1	61.3	55.7	(19)
(20)		MCWB	52.0	48.5	53.2	60.7	66.7	70.4	70.3	71.2	66.7	63.5	57.0	52.6	(20)	
(21)		2%	DB	52.2	52.7	62.3	71.6	76.9	82.8	85.5	84.5	76.6	67.4	57.8	53.2	(21)
(22)		MCWB	49.6	48.0	51.6	57.7	62.5	67.7	68.9	69.3	65.2	60.7	54.3	51.1	(22)	
(23)		5%	DB	49.5	50.2	57.6	66.3	72.9	78.9	81.8	80.4	72.1	63.7	55.2	51.1	(23)
(24)		MCWB	47.2	47.2	49.5	55.1	60.9	65.8	67.3	67.4	62.7	58.6	52.7	49.0	(24)	
(25)		10%	DB	47.1	48.2	53.8	62.0	68.7	74.8	78.3	76.3	68.0	60.7	52.8	48.6	(25)
(26)		MCWB	45.0	45.6	48.6	52.7	58.8	63.7	66.0	65.1	60.0	57.0	50.7	46.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.7	51.1	55.0	62.0	68.6	72.2	72.9	73.5	68.4	64.3	57.8	53.1	(27)
(28)		MADB	54.2	54.3	62.9	74.3	79.4	84.2	86.5	87.0	78.4	71.5	60.3	54.5	(28)	
(29)		2%	WB	50.1	49.1	53.0	58.5	64.7	69.1	70.6	70.6	66.0	61.5	54.8	51.3	(29)
(30)		MADB	51.9	51.4	59.1	68.8	73.9	79.6	82.8	82.0	74.8	66.2	57.1	53.0	(30)	
(31)		5%	WB	47.5	47.4	51.1	55.8	62.0	66.9	68.6	68.3	63.7	59.4	52.9	49.1	(31)
(32)		MADB	49.3	50.0	56.3	64.7	70.2	76.5	79.6	78.2	70.3	62.8	54.9	50.8	(32)	
(33)		10%	WB	45.1	45.5	49.1	53.4	59.7	64.8	66.7	66.1	61.4	57.3	50.9	47.0	(33)
(34)		MADB	46.9	48.1	53.3	60.3	67.2	73.5	76.3	74.2	66.7	60.1	52.8	48.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.0	13.2	17.8	21.8	21.0	21.3	21.1	21.6	20.6	16.7	11.0	9.7	(35)	
(36)		5% DB	MADB	11.0	16.8	27.6	31.5	30.7	30.1	29.3	30.7	28.5	21.1	13.1	9.9	(36)
(37)		MCWBR	9.6	12.7	18.0	18.7	17.6	16.4	14.2	16.2	17.4	14.5	10.4	8.7	(37)	
(38)		5% WB	MADB	10.5	14.3	21.6	28.1	26.2	26.6	26.0	27.7	24.8	18.6	11.4	9.6	(38)
(39)		MCWBR	9.5	11.4	14.9	17.2	15.7	15.1	13.1	15.3	15.6	13.3	9.6	8.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.338	0.370	0.400	0.411	0.410	0.409	0.403	0.372	0.357	0.331	0.311	(40)	
(41)		taud	2.425	2.381	2.296	2.232	2.238	2.267	2.266	2.309	2.376	2.414	2.423	2.397	(41)	
(42)		Ebn at Noon	232	254	263	267	269	270	268	264	260	243	222	214	(42)	
(43)		Edh at Noon	20	27	34	40	42	41	41	37	31	25	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	254	482	860	1342	1585	1754	1669	1426	1087	628	294	203	(44)	
(45)		RadStd	40	89	119	166	174	162	173	159	113	54	34	35	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page